canjob

Intelligent Candidate Discovery & Ranking

A JD-faithful, CPU-only, fully offline ranking engine that scores 100,000 candidates in ~30 seconds and explains every pick with profile evidence.

Problem

Data & AI Challenge:
Intelligent Candidate Discovery

Team

canjob

Team Leader

Navpreet Singh

01 SOLUTION OVERVIEW

What we built and why it is different

- **Solution:** an ensemble ranker that turns a JD into config, then scores every candidate with three independent lenses and fuses them.
- **Reads profiles, not just keywords:** a deterministic JD-faithful rules model encodes must-haves and the disqualifiers the JD spells out.
- **Behavior-aware:** the 23 Redrob signals modulate score so unreachable perfect-on-paper profiles sink.
- **Honest by design:** logically-impossible honeypots are detected and removed; every row carries evidence-based reasoning.

3 lenses -> 1 rank

Rules + lexical TF-IDF + MiniLM semantic, fused with Reciprocal Rank Fusion.

No LLM at rank time

Differentiator vs GPT-per-candidate systems that cannot scale to a 200k pool in production.

Config-driven

One codebase serves many jobs; each JD is its own versioned config folder.

Key requirements and the signals that matter

Extracted from the JD

- Embeddings & dense/hybrid retrieval; vector DBs (Pinecone, FAISS, Qdrant)
- Ranking / recsys + evaluation literacy (NDCG, MRR, MAP, A/B)
- LLM fine-tuning / RAG, production Python at scale
- Product-company builder profile, 5-9 yrs, founding-team mindset

Beyond keyword matching

- **Semantic fit:** MiniLM matches meaning so a recommender-systems builder ranks even without the buzzwords.
- **Availability:** response rate, recency, open-to-work, interview/offer reliability, recruiter demand.
- **Trust:** endorsements + skill duration discount lazy keyword stuffing.
- **Disqualifiers:** off-domain title with stuffed AI skills, framework-only, title-chaser tenures.

03 RANKING METHODOLOGY

Retrieve, score, rank, fuse

Rules (deterministic)

JD-faithful score from config: skill facets, experience band, product vs services, location. Minus explicit penalties. x behavioral modifier.

Lexical (TF-IDF)

Several focused queries (retrieval, ranking/eval, full JD) for keyword recall instead of one blurry mega-query.

Semantic (MiniLM)

Offline sentence embeddings, precomputed once. Cosine-to-JD captures meaning the other lenses miss.

Fusion: Reciprocal Rank Fusion (RRF)

Each lens produces a ranking; fused score = sum of weight / (k + rank). Combining on ranks, not raw scores, means no single lens scale can dominate. Weights live in jd_facets.json. Honeypots are excluded; ties break by candidate_id; scores are non-increasing.

04 EXPLAINABILITY & DATA VALIDATION

Grounded reasoning, no hallucination, suspicious profiles

- **Evidence-based:** reasoning cites the candidate's own named skills (proficiency + months), employers, and evaluation terms found in their text.
- **Mapped to the JD:** each clause connects a concrete fact to a specific JD requirement, not generic praise.
- **Honest concerns:** gaps (services-only career, framework-only, long notice) are stated, and tone matches the rank.
- **No hallucination:** every claim is pulled from the profile; nothing is invented, so Stage-4 checks pass.

0 in top 100

Honeypots (impossible profiles: expert-but-0-months, tenure > experience, role longer than career) are detected and removed. Threshold for disqualification is >10%.

High precision

We only flag logically-impossible profiles, not noisy heuristics, so legitimate candidates are never wrongly dropped.

05 END-TO-END WORKFLOW

From JD to ranked CSV

Offline boundary: only Precompute may use the network (one-time model download) and may exceed 5 min.

The Rank step that produces the CSV is CPU-only, offline, and finishes in ~30s for the full 100k pool.

06 SYSTEM ARCHITECTURE

Components and data flow

INPUTS (offline)

candidates.jsonl

100k profiles, each with 23 behavioral signals; managed via the web UI.

per-job config

JD facets, disqualifier filters and fusion weights; one folder per JD.

semantic_scores.npz

0.6 MB MiniLM cosine-to-JD scores, precomputed once and committed.

RANKING ENGINE (CPU, ~30s)

featurize

one vectorized pandas/numpy pass: profile features + 23-signal availability modifier; flags impossible honeypots.

3 scorers

rules (JD-faithful) | TF-IDF (lexical) | MiniLM (semantic).

RRF fusion

rank-based merge of the three scorers; honeypots excluded.

OUTPUT + ACCESS

top-100 CSV

candidate_id, rank, 0-1 fit score, and evidence-based reasoning. Validator-clean, UTF-8.

run it anywhere

rank.py one command | Docker (--network none) | Colab sandbox | FastAPI web UI to upload JDs, manage candidates, and run matches.

Data flow: profiles + JD config -> one vectorized featurization (+ honeypot flags) -> three scorers -> RRF -> honeypot-filtered top 100 with evidence.

Quality and constraint compliance

~30s

full 100k ranked on CPU (Docker ~50s)

0

honeypots in the submitted top 100

0.6 MB

committed semantic artifact (git-friendly)

offline

verified with docker --network none

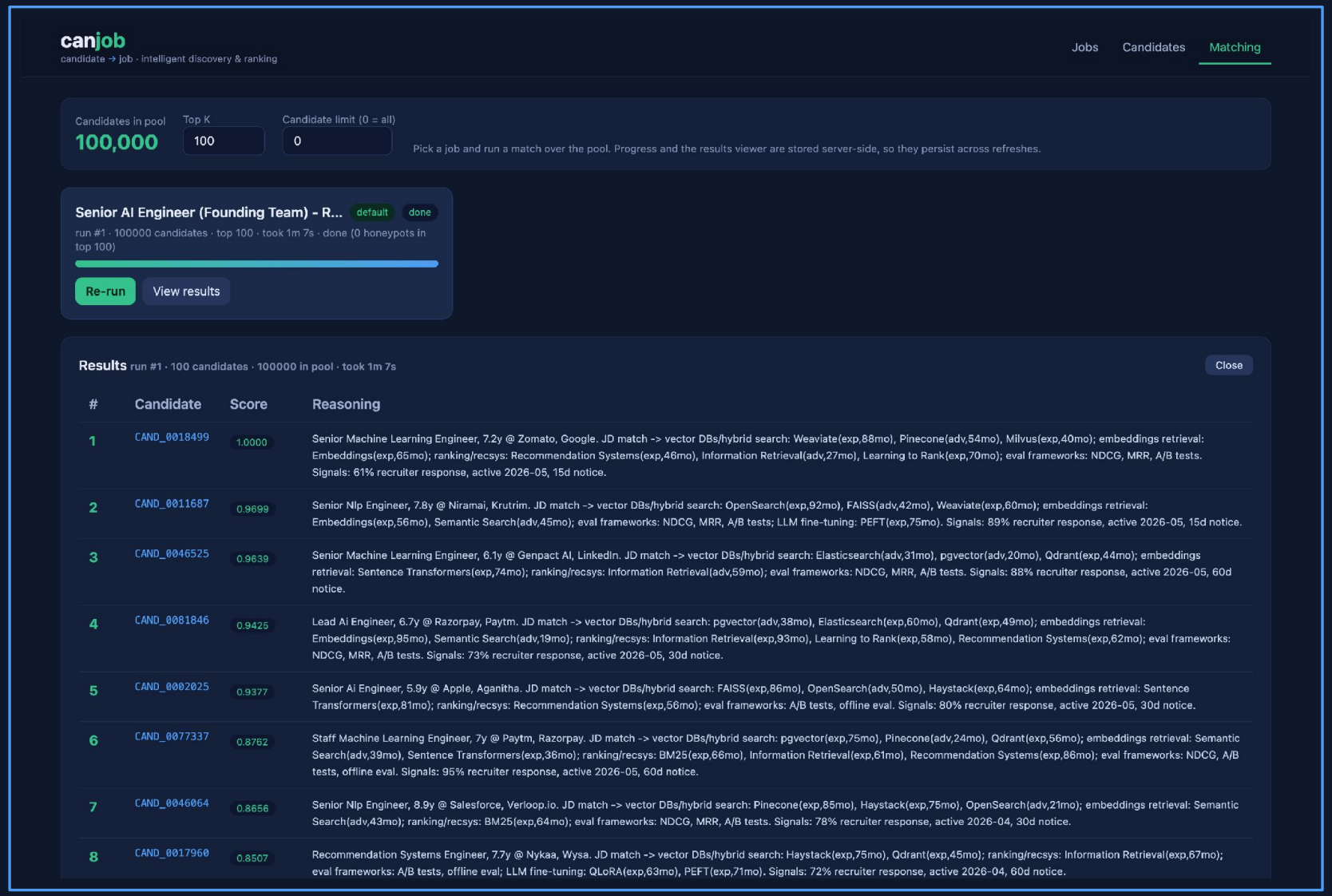
Constraint compliance

- Runtime <= 5 min: rank step ~30s for 100k
- Memory <= 16 GB: single vectorized pass
- CPU only / network off: no GPU, no API calls
- Disk <= 5 GB: artifact 0.6 MB, optional cache ~150 MB

Confidence proxy (no ground truth)

- Scored on NDCG@10 (.50), NDCG@50 (.30), MAP (.15), P@10 (.05)
- Independent lenses cross-validate: a strong consensus core ranks in every strategy
- Agreement is reported by the ranker as a defensibility signal

Drive the same engine from a browser



- **Upload a JD:** default JD pre-loaded and rendered as markdown; add ad-hoc jobs.
- **Manage the pool:** 100k candidates pre-loaded; add, remove, search, click for full profile.
- **Run with live progress:** background runs; the progress bar and results survive a refresh.
- **Same engine:** identical featurize -> 3 lenses -> RRF -> honeypot filter as rank.py.

100k pool ranked end-to-end, 0 honeypots in the top 100.

09 TECHNOLOGIES USED

And why each was chosen

pandas / numpy

vectorized featurization and scoring; no per-row loops, so 100k fits the budget.

scikit-learn

TF-IDF lexical recall, battle-tested and fast on CPU.

MiniLM (transformers)

compact offline sentence embeddings for semantic fit, precomputed once.

Reciprocal Rank Fusion

scale-free combination of heterogeneous lenses.

Docker

exact Stage-3 parity: CPU-only, offline reproduction in one command.

Google Colab

zero-setup hosted sandbox that runs end-to-end on a sample.

10 SUBMISSION ASSETS

Everything needed to reproduce and verify

- **GitHub:** github.com/NavpreetDevpuri/canjob (public, real iterative history)
- **Reproduce:** python rank.py --candidates ./candidates.jsonl --out ./submission.csv
- **Sandbox:** Google Colab run-all notebook with a config panel and bundled sample
- **Docker:** docker build -t canjob . then docker run --network none ...
- **Web UI (bonus):** FastAPI + single-page app: upload JDs, manage candidates, run matches with live progress
- **Submission CSV:** top 100, UTF-8, candidate_id,rank,score,reasoning (validator-clean)
- **Metadata:** submission_metadata.yaml mirrors the portal fields

Thank you

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